

Question	Marking Guidance	Mark	Comments
04.1	69.3 cm ³ solvent, 29.7 cm ³ water, 1.0 cm ³ acid (box 2);	1	
04.2	1. Temperature; 2. Agitation/mixing/stirring; 3. Source/age/type of blueberries; 4. Crushing of the blueberries; 5. Rinsing of the blueberries prior to mixing; 6. Concentration of ethanol/acid;	2 max	Do not accept pH. Accept 'Filtering method'.
04.3	1. Higher absorbance indicates more anthocyanin OR Higher absorbance indicates more membrane damage/permeability OR (G not zero because) some anthocyanin released when blueberries are crushed OR (G not zero because) some membrane damage when blueberries are crushed; 2. More membrane damage/permeability results in more anthocyanin release 3. (E and F greater than water because) phospholipids dissolve in ethanol; 4. (E greater than F because) acid denatures membrane proteins;	4	For 'anthocyanin' accept 'pigment'. 1. A direct comparative statement is not needed, can be taken from the answer as a whole. 1 and 2. Accept 'most' for 'more'. 4. Accept description of denaturation in terms of change in tertiary structure or breaking of hydrogen/ionic bonds.
04.4	1. Use known concentration of blueberry juice/extract OR Use known concentration of anthocyanin/pigment (solution) OR Use known concentration of (extraction) solvent to be added to blueberries; 2. Prepare dilution series; 3. Compare (results) with colour standards to give score/value/concentration;	3	2. Accept descriptions and 'serial dilutions' 2. Accept dilution series in terms of pigment or solvent. 3. For 'colour standards' accept 'dilutions'.

